CroDoSR: Tensor Cross-Domain Rank for Hyperspectral Image Super-Resolution

Zhong-Cheng Wu, Ya-Jun Li, Ting-Zhu Huang, *Member, IEEE*, Liang-Jian Deng, *Senior Member, IEEE*, and Gemine Vivone, *Senior Member, IEEE*

***Abstract*— Hyperspectral image super-resolution (HSI SR) aims to combine the detailed spectral information of hyperspectral images with the spatial resolution of multispectral images, thus enhancing the ability to extract valuable insights across various applications. Recently, the tensor singular value decomposition (*t*-SVD) has emerged as a powerful tool and has been introduced into the HSI SR field for exploring low-rank prior information. For *t*-SVD, the domain transform is crucial to acquiring more low-rank data characteristics. Nevertheless, previous efforts on domain transform have only involved the single transformed domain (i.e., *single domain*), while ignoring the potential pursuing the lower rankness in multiple successional transformed domains, termed cross-domain (CD). In this article, we propose a novel CD-based *t*-SVD and define the corresponding tensor CD rank based on a pivotal observation, i.e., the low-rank behavior of HSI in CD is more significant than that in single domain. More specifically, we first define a successional linear transform (SLT) to establish the CD concept, then develop a novel CD-based *t*-SVD and tensor CD rank, and theoretically deduce a new tensor CD-nuclear norm as the convex approximation of CD rank. Equipped with such a CD rank, we thus formulate a CD-rank-constrained minimization model for the HSI SR task, called CroDoSR, which is effectively solved by the alternating direction method of multipliers (ADMMs). Comprehensive experiments on several widely used datasets evidently demonstrate the superiority of the proposed CroDoSR method.**

***Index Terms*— Cross domain (CD), hyperspectral image super-resolution (HSI SR), image fusion, remote sensing, tensor CD-nuclear norm, tensor CD rank, tensor singular value decomposition (*t*-SVD).**

I. Introduction

HIGH spatial-resolution hyperspectral images (HR-HSIs) are very important in computer vision and significantly contribute to many high-level applications, e.g., classification [2], [3], target detection [4], [5], and face recognition [6], [7]. Due to the inherent limitations of hardware systems, current hyperspectral imaging sensors can only capture lower spatial-resolution HSIs (LR-HSIs), which generally provide richer spectral information but inferior spatial details. However, such low spatial resolution will inevitably deteriorate the performance of subsequent image processing applications. Conversely, multispectral imaging sensors can acquire higher spatial-resolution images (HR-MSIs), even if they suffer from lower spectral resolution. To overcome the hardware limitations, hyperspectral image super-resolution (HSI SR) is naturally developed as an algorithmic tool to enhance the spatial resolution of the LR-HSIs by leveraging the HR-MSIs, thus generating the underlying HR-HSIs.

In the past decades, HSI SR has received considerable attention, and many excellent works are developed. Mainly, they can be divided into three categories, including matrix factorization (MF)-based methods, tensor factorization (TF)-based methods, and deep learning (DL)-based methods. The MF-based methods, see [8], [9], [10], [11], typically unfold the underlying HR-HSI along with the spectral dimension into a matrix, then decompose it as a basis multiplied by a coefficient. In this case, the HSI SR problem is converted to the estimation of the basis and coefficient. For instance, Yokoya et al. [9] perform the SR by nonnegative MF. In general, the nonnegative MF is not numerically unique, thus leading to unstable even unsatisfactory results. Furthermore, some methods, such as [10] and [11], instead alternately update the basis and coefficient, expecting to realize more accurate results. However, two drawbacks of the MF-based methods are still reflected: 1) only the spectral correlation is considered and 2) the inherent image structure is impaired by the unfolding operation.

Recently, the DL-based methods, see [12], [13], [14], [15], [16], [17], [18], have been rapidly developed for the HSI SR problem. For instance, Yang et al. [13] propose two deep convolutional neural network (CNN) branches concatenated with fully connected layers. Moreover, manifold structure learning [12], unfolding architecture [14], and two-stream regularization [19] are also proposed. Unfortunately, these methods fail to consider any specific image prior, thus resulting in the training-data-dependent deviation on testing datasets. Subsequently, many unsupervised DL methods, see [18], [20], [21], [22], actively contribute to improving data generalization. Nevertheless, DL methods typically suffer from inferior physical interpretability, especially in parameter

Received 10 May 2024; revised 30 July 2024 and 4 September 2024; accepted 5 September 2024. Date of publication 10 September 2024; date of current version 24 September 2024. This work was supported in part by Natural Science Foundation of Sichuan Province under Grant 2024NSFSC0038 and in part by National Natural Science Foundation of China under Grant 12171072 and Grant 12271083. *(Corresponding authors: Ting-Zhu Huang; Liang-Jian Deng.)*

Zhong-Cheng Wu, Ya-Jun Li, Ting-Zhu Huang, and Liang-Jian Deng are with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: wuzhch97@163.com; liyajun1235@163.com; tingzhuhuang@126.com; liangjian.deng@uestc.edu.cn).

Gemine Vivone is with the Institute of Methodologies for Environmental Analysis, National Research Council, 85050 Tito, Italy, and also with the National Biodiversity Future Center (NBFC), 90133 Palermo, Italy (e-mail: gemine.vivone@imaa.cnr.it).

Digital Object Identifier 10.1109/TGRS.2024.3457673

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

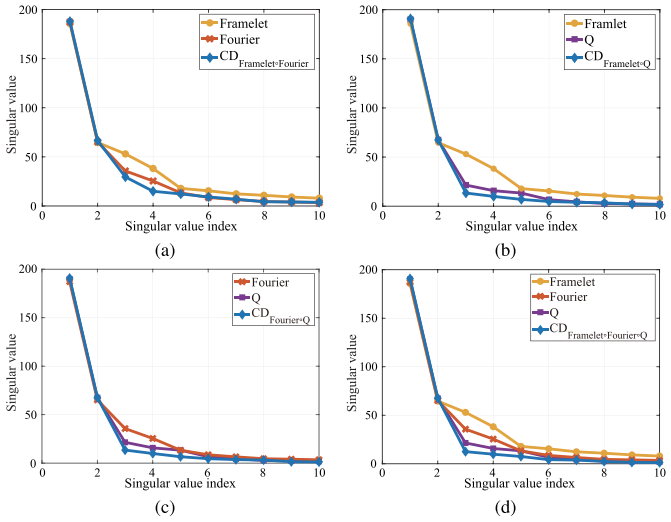


Fig. 1. Tensor singular curves for an HSI (source: PaU dataset) on the single domain (i.e., Fourier domain, framelet domain, and Q-domain [1]) and the CD. (a) Fourier + framelet, (b) Q + framelet, (c) Q + Fourier, and (d) Q + Fourier + framelet. From the comparison, the CD clearly contributes the faster decreasing for singular values than the single domain, thus verifying the potential of the former in characterizing the global low rankness of HSIs.

learning, compared with traditional prior-based optimization methods.

The TF-based methods, see [23], [24], [25], [26], [27], [28], [29], [30], [31], have been extensively and fully investigated in the HSI SR problem. Some of these methods, see [23], [24], [26], construct an HSI SR model by Tucker decomposition and transfer the SR problem into the estimation of a core tensor and three dictionaries. In addition, Xu et al. [27] develop an HSI SR model by assuming that the relationship between LR-HSI and HR-MSI can be characterized by coupled CANDECOMP/PARAFAC (CP) decomposition. Furthermore, tensor singular value decomposition (t-SVD) has also been introduced for the HSI SR problem. By employing the t-SVD-based low-rank regularizer, Dian and Li [31] formulate a more efficient SR modeling. For t-SVD, the domain transform is crucial to acquire more low-rank data characteristics. Nevertheless, previous efforts on domain transform have only involved the single domain, e.g., Fourier [32] and framelet [33], while ignoring the potential to pursue the lower rankness in multiple successional transformed domains, thus inevitably leading to limited SR performance.

In this article, we first define the concepts of successional linear transform (SLT) and cross domain (CD), then develop CD-oriented tensor–tensor product (t-product), t-SVD, tensor CD rank, and tensor CD-nuclear norm. Afterward, we propose a tensor CD-rank minimization model for the HSI SR problem, called CroDoSR, aiming to overcome the limitations of the existing t-SVD-based methods. Compared with the previous methods, the CroDoSR can more significantly characterize the low-rankness behavior of HSIs by SLT as prior information, thus allowing for more promising SR performance. To optimize the CroDoSR model, we design an efficient alternating direction method of multiplier (ADMM)-based solving algorithm. Extensive experiments on several datasets demonstrate the superiority of the proposed CroDoSR.

The *contributions* of this article are summarized as follows.

1) We investigate the potential of SLT for t-SVD and develop several fundamental concepts, e.g., SLT and CD, which markedly improve the existing t-SVD theories. Especially, we also provide some novel definitions, including the CD-oriented t-product, t-SVD, and tensor CD rank.
2) We theoretically deduce a new tensor nuclear norm (TNN) as the convex relaxation of tensor CD rank, named CD-nuclear norm. By embedding the tensor CD-nuclear norm, we further formulate a tensor CD-rank minimization model for the HSI SR problem, called CroDoSR.
3) Comprehensive experiments validate the superiority of the CroDoSR over other SR methods, especially those single-domain-based ones, both qualitatively and quantitatively.

The overall architecture of the proposed CroDoSR method is visualized in Fig. 2 for better comprehension. The remainder of this article is organized as follows. The related work and motivation are analyzed in Section II. The notation is introduced in Section III. The CD rank and CD-nuclear norm are presented in Section IV. The CroDoSR model is formulated in Section V. The numerical experiments are performed in Section VI. Finally, the conclusion is described in Section VII.

II. Related Works

The HSI elements are highly correlated with each other, which implies that the image tensor is approximately low rank [34], [35]. The commonly used definitions of tensor rank are all related to specific tensor decompositions [36]. Based on CP decomposition, the CP rank is defined as the smallest number of rank-one tensors. However, optimizing the CP rank is NP-hard. To overcome this challenge, Liu et al. [37] define a Tucker rank-based TNN, named SNN. However, either Tucker rank or SNN requires matrix unfolding, which destroys the inherent structure of the tensor, thus resulting in performance degradation.

Based on the discrete Fourier transform (DFT), Kilmer and Martin [32] propose the t-product and t-SVD, then giving a definition of the TNN. Afterward, Kernfeld et al. [38] generalize the t-product by introducing the discrete cosine transform (DCT). Following [38], Lu et al. [39] further propose a new TNN based on any invertible linear transform. When a specific transformation is prescribed, a type of TNN can be generated. For instance, Jiang et al. [33] propose a framelet transform-based TNN. Furthermore, Kong et al. [1] define the Q-domain and propose the tensor Q-rank with a data-dependent transform. Although these t-SVD-based methods have yielded promising results, they are all implemented in single domains while ignoring the potential in successional domains, which provides the vital motivation of this article.

Motivation: To fully characterize the low rankness of HSIs, we investigate the tensor singular value curves[1] of HSI on

[1]Here, the tensor singular value curve is obtained by plotting the diagonal element vector of $\arg\max_{\mathbf{S}^{(i)}} \|\mathbf{S}^{(i)}\|_0$, where $\mathbf{S}^{(i)}$ is the ith frontal slice of $\mathcal{S}$ generated by t-SVD, and $\|\cdot\|_0$ indicates ℓ_0 norm for counting nonzero elements. Please see Section III and Theorem 1 for more details.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

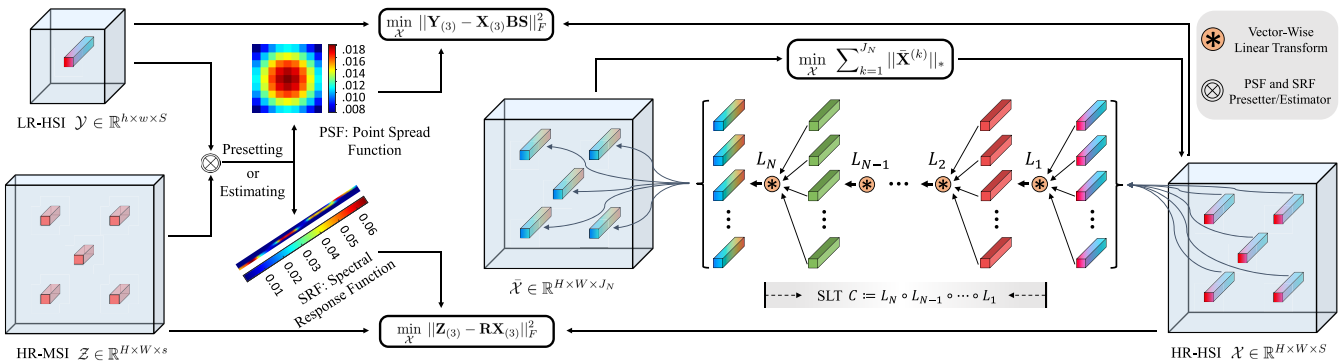


Fig. 2. Architecture of the proposed CroDoSR method.

both single domain and CD (i.e., successional transformed domains), as shown in Fig. 1. From the comparison, the CD clearly contributes the faster decreasing for singular values than the single domain, verifying the potential of the former in capturing the global low rankness of HSIs.

III. Notations

Generally, scalars, vectors, matrices, and tensors are denoted by lowercase letters, e.g., x, lowercase bold letters, e.g., $\mathbf{x}$, uppercase bold letters, e.g., $\mathbf{X}$, and calligraphic letters, e.g., $\mathcal{X}$, respectively. For an Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$, we can express its $(i_1, i_2, \ldots, i_N)$th entry as $\mathcal{X}(i_1, i_2, \ldots, i_N)$ or $x_{i_1,i_2,\ldots,i_N}$. The mode-k matricization of $\mathcal{X}$ is defined as $\mathbf{X}_{(k)} \in \mathbb{R}^{I_k\times\prod_{j\neq k} I_j}$, whose elements satisfy $\mathbf{X}_{(k)}(i_k, \overline{i_1\cdots i_{k-1}i_{k+1}\cdots i_N}) = \mathcal{X}(i_1, i_2, \ldots, i_N)$ with $\overline{i_1\cdots i_{k-1}i_{k+1}\cdots i_N} = 1 + \sum_{n=1,n\neq k}^{N}(i_n - 1)\prod_{j=1,j\neq k}^{n-1} I_j$. Given a matrix $\mathbf{Y} \in \mathbb{R}^{J\times I_k}$; then, the k-mode multiplication between $\mathcal{X}$ and $\mathbf{Y}$ yields a tensor $\mathcal{Z} = \mathcal{X} \times_k \mathbf{Y} \in \mathbb{R}^{I_1\times\cdots\times I_{k-1}\times J\times I_{k+1}\times\cdots\times I_N}$, which numerically requires $z_{i_1,\ldots,i_{k-1},j,i_{k+1},\ldots,i_N} = \sum_{i_k} x_{i_1,\ldots,i_k,\ldots,i_N} y_{j,i_k}$. When $N = 3$, the kth frontal slice of $\mathcal{X}$ is denoted by $\mathcal{X}(:, :, k)$ or, more compactly, $\mathbf{X}^{(k)}$. Similarly, the (i, j)th tube is denoted by $\mathcal{X}(i, j, :)$. The inner product between $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3}$ and $\mathcal{Y} \in \mathbb{R}^{I_1\times I_2\times I_3}$ is defined as $\langle\mathcal{X}, \mathcal{Y}\rangle := \sum_{i,j,k} x_{i,j,k} y_{i,j,k}$. Then, the Frobenius norm can be defined as $||\mathcal{X}||_F := (\langle\mathcal{X}, \mathcal{X}\rangle)^{1/2}$. For a matrix $\mathbf{X}$, the spectral norm is defined as $||\mathbf{X}||_\sigma := \max_i \delta_i(\mathbf{X})$, and the matrix nuclear norm is defined as $||\mathbf{X}||_* := \sum_i \delta_i(\mathbf{X})$, where $\delta_i(\mathbf{X})$ represents the ith matrix singular value of $\mathbf{X}$.

IV. Tensor CD Rank and CD-Nuclear Norm

In this section, we develop some basic definitions and theories, including SLT, CD, CD rank, and CD-nuclear norm.

Definition 1 [SLT]: Given a third-order tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times J_0}$ and N ($N \geq 1$) linear transforms $L_i : \mathbb{C}^{I_1\times I_2\times J_{i-1}} \to \mathbb{C}^{I_1\times I_2\times J_i}$, $i = 1, 2, \ldots, N$, then the SLT is defined as the function composition of these N transforms, i.e., $C := L_N \circ L_{N-1} \circ \cdots \circ L_i \circ \cdots \circ L_1 : \mathbb{R}^{I_1\times I_2\times J_0} \to \mathbb{C}^{I_1\times I_2\times J_N}$, which mathematically allows

$$\begin{aligned} C(\mathcal{X}) &= (L_N \circ L_{N-1} \circ \cdots \circ L_i \circ \cdots \circ L_1)(\mathcal{X}) \\ &= \mathcal{X} \times_3 \mathbf{L}_1 \times_3 \mathbf{L}_2 \times_3 \cdots \times_3 \mathbf{L}_N \end{aligned} \tag{1}$$

where $\mathbf{L}_i \in \mathbb{C}^{J_i\times J_{i-1}}$, $i = 1, 2, \ldots, N$, indicates the corresponding transform matrix of the ith given linear transform.[2]

Remark 1: According to Definition 1, the SLT is essentially a recoverable linear transform, which is composited from multiple existing transforms. Namely, any existing transform L_i, e.g., framelet and DFT, can be encompassed in SLT as a special case with $N = 1$. Interestingly, for different collections $\{L_i\}_{i=1}^N$, the SLT can exhibit different composite properties, which explicitly generates a new nonexisting transform C, thus contributing to even more significant low-rankness prior, as shown in Fig. 1.

Definition 2 [Cross-Domain (CD)]: For a given SLT $C = L_N \circ L_{N-1} \circ \cdots \circ L_i \circ \cdots \circ L_1$, where $L_i : \mathbb{C}^{I_1\times I_2\times J_{i-1}} \to \mathbb{C}^{I_1\times I_2\times J_i}$, $i = 1, 2, \ldots, N$, are N ($N \geq 1$) linear transforms, then the CD is defined as the collection $\{\bar{\mathcal{X}} \in \mathbb{C}^{I_1\times I_2\times J_N} : \bar{\mathcal{X}} = C(\mathcal{X}),\ \mathcal{X} \in \mathbb{R}^{I_1\times I_2\times J_0}\}$. For simplicity, the CD is typically denoted by $\mathrm{CD}_{L_N\circ L_{N-1}\circ\cdots\circ L_1}$ throughout this article.

Following Definition 2, we further deduce the CD-oriented concepts. More preliminaries can be referenced in [38] and [40].

A. Tensor CD Rank

For a given third-order tensor $\bar{\mathcal{X}} \in \mathbb{R}^{I_1\times I_2\times J_N}$, we first denote $\bar{\mathbf{X}}$ as the block-diagonal matrix of $\bar{\mathcal{X}}$, i.e.,

$$\bar{\mathbf{X}} = \mathtt{bdiag}(\bar{\mathcal{X}}) = \begin{bmatrix} \bar{\mathbf{X}}^{(1)} & & & \\ & \bar{\mathbf{X}}^{(2)} & & \\ & & \ddots & \\ & & & \bar{\mathbf{X}}^{(J_N)} \end{bmatrix} \tag{2}$$

where $\mathtt{bdiag}$ is an operator mapping $\bar{\mathcal{X}}$ to $\bar{\mathbf{X}} \in \mathbb{R}^{I_1 J_N\times I_2 J_N}$.

Definition 3 (CD-Oriented t-Product): For any SLT C in Definition 1, the C-based t-product between tensors $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times J_0}$ and $\mathcal{Y} \in \mathbb{R}^{I_2\times I_3\times J_0}$ is defined as follows:

$$\mathcal{X} *_C \mathcal{Y} = C^{-1}\left(\bar{\mathcal{X}} \odot \bar{\mathcal{Y}}\right) \tag{3}$$

where $\bar{\mathcal{X}} = C(\mathcal{X})$, $\bar{\mathcal{Y}} = C(\mathcal{Y})$, and $\odot$ is the frontal-slicewise product, i.e., $\mathcal{Z} = \bar{\mathcal{X}} \odot \bar{\mathcal{Y}} \Leftrightarrow \mathbf{Z}^{(i)} = \bar{\mathbf{X}}^{(i)}\bar{\mathbf{Y}}^{(i)}$ for $i = 1, 2, \ldots, J_N$.

[2]For the ith linear transform $L_i : \mathbb{C}^{I_1\times I_2\times J_{i-1}} \to \mathbb{C}^{I_1\times I_2\times J_i}$, $J_{i-1} \neq J_i$ may be satisfied, i.e., L_i is not invertible. However, L_i is required to be recoverable, i.e., $\mathcal{X} = (L_i^* \circ L_i)(\mathcal{X})$. For brevity, we can denote $L_i^{-1} := L_i^*$ for such a case.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

Definition 4 (Conjugate Transpose [38]): Given a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$, its conjugate transpose $\mathcal{X}^{\mathrm{T}} \in \mathbb{R}^{I_2 \times I_1 \times J_0}$ under CD satisfies $\bar{\mathcal{X}}^{\mathrm{T}}(:,:,i) = \bar{\mathcal{X}}(:,:,i)^{\mathrm{T}}$ for $i = 1, 2, \ldots, J_N$.

Definition 5 (Identity Tensor [38]): For a given SLT C, tensor $\mathcal{I} \in \mathbb{R}^{I_1 \times I_1 \times J_0}$ is called an identity tensor if all frontal slices of $\bar{\mathcal{I}} = C(\mathcal{I})$ are $I_1 \times I_1$ identity matrices.

Definition 6 (Orthogonal Tensor [38]): Given an SLT C, tensor $\mathcal{Q} \in \mathbb{R}^{I_1 \times I_1 \times J_0}$ is orthogonal if $\mathcal{Q}^{\mathrm{T}} *_C \mathcal{Q} = \mathcal{Q} *_C \mathcal{Q}^{\mathrm{T}} = \mathcal{I}$ can be satisfied.

Theorem 1 (t-SVD [38]): For any SLT C given in Definition 1, $*_C$-based t-SVD of tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$ is given by

$$\mathcal{X} = \mathcal{U} *_C \mathcal{S} *_C \mathcal{V}^{\mathrm{T}} \tag{4}$$

where $\mathcal{U} \in \mathbb{R}^{I_1 \times I_1 \times J_0}$ and $\mathcal{V} \in \mathbb{R}^{I_2 \times I_2 \times J_0}$ are orthogonal, and $\mathcal{S} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$ is an f-diagonal tensor whose each frontal slice is a diagonal matrix.

Definition 7 (Tensor CD Rank): Given an SLT C, the tensor CD rank of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$, denoted as $\mathrm{rank}_{\mathrm{CD}}(\mathcal{X})$, is defined to be the number of nonzero singular tubes of $\mathcal{S}$, where $\mathcal{S}$ is from the t-SVD $\mathcal{X} = \mathcal{U} *_C \mathcal{S} *_C \mathcal{V}^{\mathrm{T}}$. That is,

$$\mathrm{rank}_{\mathrm{CD}}(\mathcal{X}) := \#\{i : \mathcal{S}(i,i,:) \neq \mathbf{0}\}. \tag{5}$$

Since $\mathrm{rank}_{\mathrm{CD}}(\mathcal{X})$ is nonconvex and even discrete, its convex relaxation, i.e., the CD-nuclear norm, is further deduced.

B. Tensor CD-Nuclear Norm

Analogously to the convex approximation in [32], the tensor CD-nuclear norm can be developed from the spectral norm.

Definition 8 (Tensor Spectral Norm): For a given SLT C, C-based tensor spectral norm of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$, denoted by $||\mathcal{X}||_\sigma$, is defined by

$$||\mathcal{X}||_\sigma := \left|\left|\bar{\mathbf{X}}\right|\right|_\sigma \tag{6}$$

where $\bar{\mathbf{X}} = \mathtt{bdiag}(\bar{\mathcal{X}}) \in \mathbb{R}^{I_1 J_N \times I_2 J_N}$ with $\bar{\mathcal{X}} = C(\mathcal{X}) \in \mathbb{R}^{I_1 \times I_2 \times J_N}$.

Subsequently, by Definition 8, we can deduce a tensor CD-rank-customized nuclear norm by the dual norm of the tensor spectral norm. Let $C = L_N \circ L_{N-1} \circ \cdots \circ L_i \circ \cdots \circ L_1$ $(N \geq 1)$, where the ith linear transform L_i relates to the ith transform matrix $\mathbf{L}_i$. Assume that

$$\mathbf{L}_i^{\mathrm{T}} \mathbf{L}_i = \mathbf{L}_i \mathbf{L}_i^{\mathrm{T}} = \tau_i \mathbf{I}, \quad i = 1, 2, \ldots, N \tag{7}$$

where $\tau_i > 0$, $i = 1, 2, \ldots, N$, are the specific constants. Accordingly, we can obtain

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \frac{1}{\prod_{i=1}^N \tau_i} \langle \bar{\mathbf{X}}, \bar{\mathbf{Y}} \rangle \tag{8}$$

and

$$||\mathcal{X}||_F = \frac{1}{\sqrt{\prod_{i=1}^N \tau_i}} \left|\left|\bar{\mathbf{X}}\right|\right|_F. \tag{9}$$

For any $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$ and $\mathbf{Z} \in \mathbb{R}^{I_1 J_N \times I_2 J_N}$, we give

$$\begin{aligned} ||\mathcal{X}||_{*,\mathrm{CD}} &:= \sup_{||\mathcal{Y}||_\sigma \leq 1} \langle \mathcal{X}, \mathcal{Y} \rangle = \sup_{||\bar{\mathbf{Y}}||_\sigma \leq 1} \frac{1}{\prod_{i=1}^N \tau_i} \langle \bar{\mathbf{X}}, \bar{\mathbf{Y}} \rangle \\ &\leq \sup_{||\mathbf{Z}||_\sigma \leq 1} \frac{1}{\prod_{i=1}^N \tau_i} \langle \bar{\mathbf{X}}, \mathbf{Z} \rangle \\ &= \frac{1}{\prod_{i=1}^N \tau_i} \left|\left|\bar{\mathbf{X}}\right|\right|_* \end{aligned} \tag{10}$$

where $\bar{\mathbf{Y}} = \mathtt{bdiag}(\bar{\mathcal{Y}}) \in \mathbb{R}^{I_1 J_N \times I_2 J_N}$ is a block diagonal matrix, and $\mathbf{Z} \in \mathbb{R}^{I_1 J_N \times I_2 J_N}$ is an arbitrary matrix of the same size. Considering the t-SVD $\mathcal{X} = \mathcal{U} *_C \mathcal{S} *_C \mathcal{V}^{\mathrm{T}}$, we further have

$$\begin{aligned} ||\mathcal{X}||_{*,\mathrm{CD}} &:= \sup_{||\mathcal{Y}||_\sigma \leq 1} \langle \mathcal{X}, \mathcal{Y} \rangle \geq \langle \mathcal{U} *_C \mathcal{S} *_C \mathcal{V}^{\mathrm{T}}, \mathcal{U} *_C \mathcal{V}^{\mathrm{T}} \rangle \\ &= \langle \mathcal{U}^{\mathrm{T}} *_C \mathcal{U} *_C \mathcal{S}, \mathcal{V}^{\mathrm{T}} *_C \mathcal{V} \rangle \\ &= \langle \mathcal{S}, \mathcal{I} \rangle = \frac{1}{\prod_{i=1}^N \tau_i} \langle \bar{\mathbf{S}}, \bar{\mathbf{I}} \rangle = \frac{1}{\prod_{i=1}^N \tau_i} \mathtt{Tr}(\bar{\mathbf{S}}) \\ &= \frac{1}{\prod_{i=1}^N \tau_i} \left|\left|\bar{\mathbf{X}}\right|\right|_*. \end{aligned} \tag{11}$$

Combining (10) and (11), the tensor CD-nuclear norm, i.e., $||\mathcal{X}||_{*,\mathrm{CD}}$, can be naturally defined as follows.

Definition 9 (Tensor CD-Nuclear Norm): For any SLT C satisfying (7), the tensor CD-nuclear norm of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$, denoted by $||\mathcal{X}||_{*,\mathrm{CD}}$, is defined as follows:

$$||\mathcal{X}||_{*,\mathrm{CD}} := \langle \mathcal{S}, \mathcal{I} \rangle = \frac{1}{\prod_{i=1}^N \tau_i} \left|\left|\bar{\mathbf{X}}\right|\right|_* \tag{12}$$

where $\tau_i > 0$, $i = 1, 2, \ldots, N$, are several constants.

Correspondingly, the solution for the proximal operator of the tensor CD-nuclear norm $||\mathcal{X}||_{*,\mathrm{CD}}$ is given in Theorem 2.

Theorem 2: For a given $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$, any $\lambda > 0$, and

$$\min_{\mathcal{X}} \ \lambda ||\mathcal{X}||_{*,\mathrm{CD}} + \frac{1}{2} ||\mathcal{X} - \mathcal{Y}||_F^2 \tag{13}$$

then the solution of minimization problem (13) is denoted by

$$\mathrm{Prox}^{\lambda}_{C,||\cdot||_{*,\mathrm{CD}}}(\mathcal{Y}) := \arg\min_{\mathcal{X}} \ \lambda ||\mathcal{X}||_{*,\mathrm{CD}} + \frac{1}{2} ||\mathcal{X} - \mathcal{Y}||_F^2 \tag{14}$$

where $C : \mathbb{R}^{I_1 \times I_2 \times J_0} \to \mathbb{C}^{I_1 \times I_2 \times J_N}$ is the SLT, $||\cdot||_{*,\mathrm{CD}}$ is the tensor CD-nuclear norm, and $\mathrm{Prox}^{\lambda}_{C,||\cdot||_{*,\mathrm{CD}}}(\mathcal{Y})$ is given in Algorithm 1.

Algorithm 1 Solver of $\mathrm{Prox}^{\lambda}_{C,||\cdot||_{*,\mathrm{CD}}}(\mathcal{Y})$

Input: $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times J_0}$, λ, and SLT $C : \mathbb{R}^{I_1 \times I_2 \times J_0} \to \mathbb{C}^{I_1 \times I_2 \times J_N}$.

1: $\bar{\mathcal{Y}} = C(\mathcal{Y})$.
2: **for** $i = 1$ to J_N **do**
3: $\quad [\mathbf{U}, \mathbf{S}, \mathbf{V}^{\mathrm{T}}] = \mathtt{SVD}(\bar{\mathbf{Y}}^{(i)})$;
4: $\quad \bar{\mathbf{Y}}^{(i)} = \mathbf{U}(\mathbf{S} - \lambda)_+ \mathbf{V}^{\mathrm{T}}$ with $(\cdot)_+ = \max\{\cdot, 0\}$.
5: **end for**
6: $\mathrm{Prox}^{\lambda}_{C,||\cdot||_{*,CD}}(\mathcal{Y}) = C^{-1}(\bar{\mathcal{Y}})$.

Output: $\mathrm{Prox}^{\lambda}_{C,||\cdot||_{*,CD}}(\mathcal{Y})$.

V. Tensor CD-Rank Minimization for HSI SR

In this section, we formulate a unified paradigm for HSI SR using tensor CD-rank minimization, called CroDoSR.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

A. HSI SR Problem

Let $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ be the underlying HR-HSI with $H \times W$ spatial pixels and S spectral bands. The acquired LR-HSI and HR-MSI are, respectively, denoted by $\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$ and $\mathcal{Z} \in \mathbb{R}^{H\times W\times s}$, where $S \gg s$, $h = H/r$, and $w = W/r$ with r being the scaling factor. Typically, the LR-HSI $\mathcal{Y}$ can be regarded as the spatially degraded version of the HR-HSI $\mathcal{X}$. Mathematically, we have

$$\mathbf{Y}_{(3)} = \mathbf{X}_{(3)}\mathbf{BS} \tag{15}$$

where $\mathbf{Y}_{(3)} \in \mathbb{R}^{S\times hw}$ and $\mathbf{X}_{(3)} \in \mathbb{R}^{S\times HW}$ are the mode-3 unfoldings of $\mathcal{Y}$ and $\mathcal{X}$, respectively. Moreover, $\mathbf{B} \in \mathbb{R}^{HW\times HW}$ and $\mathbf{S} \in \mathbb{R}^{HW\times hw}$ are the blurring and downsampling matrices.

For another, the HR-MSI $\mathcal{Z}$ can be modeled as the spectrally downsampled version of the HR-HSI $\mathcal{X}$, i.e.,

$$\mathbf{Z}_{(3)} = \mathbf{R}\mathbf{X}_{(3)} \tag{16}$$

where $\mathbf{Z}_{(3)} \in \mathbb{R}^{s\times HW}$ is the mode-3 unfolding of $\mathcal{Z}$, and $\mathbf{R} \in \mathbb{R}^{s\times S}$ is the spectral response matrix of the multispectral imaging sensor. Based on (15) and (16), the general minimization model of HSI SR can be given by

$$\min_{\mathcal{X}} \ \alpha_1\left\|\mathbf{Y}_{(3)} - \mathbf{X}_{(3)}\mathbf{BS}\right\|_F^2 + \alpha_2\left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 \tag{17}$$

where $\alpha_i > 0$ and $i = 1, 2$ are the two regularization parameters.

Notably, minimization problem (17) adopts only two degeneration assumptions for reconstructing the HR-HSI $\mathcal{X}$, which contains a lot of redundant information, leading to an inadequate characterization and limited modeling. To fully explore the intrinsic properties of the HR-HSI, the t-SVD-based low multirank prior [31] is introduced into the HSI SR, yielding more promising results. Nevertheless, the low multirank regularization uses only a single transform, i.e., Fourier transform, which is clearly insufficient to better capture the global low rankness of HSIs, as illustrated in Fig. 1. Accordingly, we develop the CroDoSR model by leveraging the low CD-rank prior, aiming to further improve the potential of the SR model.

B. CroDoSR Model and Algorithm

Relying on the above analysis, by imposing the low CD-rank prior on model (17), we thus formulate the CroDoSR model as follows:

$$\min_{\mathcal{X}} \ \alpha_1\left\|\mathbf{Y}_{(3)} - \mathbf{X}_{(3)}\mathbf{BS}\right\|_F^2 + \alpha_2\left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 + \lambda \text{rank}_{\text{CD}}(\mathcal{X}) \tag{18}$$

where $\lambda > 0$ is a regularization parameter.

Since directly optimizing (18) is NP-hard, model (18) can be alternatively reformulated by using the convex approximation of $\text{rank}_{\text{CD}}(\mathcal{X})$ as follows:

$$\min_{\mathcal{X}} \ \alpha_1\left\|\mathbf{Y}_{(3)} - \mathbf{X}_{(3)}\mathbf{BS}\right\|_F^2 + \alpha_2\left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 + \lambda ||\mathcal{X}||_{*,\text{CD}} \tag{19}$$

where $||\mathcal{X}||_{*,\text{CD}}$ is the defined tensor CD-nuclear norm.

To optimize the resulting CroDoSR model (19), we develop an efficient ADMM-based solving algorithm. By introducing the auxiliary variable $\mathcal{V} = \mathcal{X}$, model (19) is first transformed into the following constrained problem:

$$\begin{aligned}\min_{\mathcal{X},\mathcal{V}} \ & \alpha_1\left\|\mathbf{Y}_{(3)} - \mathbf{X}_{(3)}\mathbf{BS}\right\|_F^2 + \alpha_2\left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 \\ & + \lambda ||\mathcal{V}||_{*,\text{CD}} \\ \text{s.t. } & \mathcal{V} = \mathcal{X}.\end{aligned} \tag{20}$$

The augmented Lagrangian function of (20) is

$$\begin{aligned}\mathcal{L}(\mathcal{X}, \mathcal{V}, \Theta) = {} & \alpha_1\left\|\mathbf{Y}_{(3)} - \mathbf{X}_{(3)}\mathbf{BS}\right\|_F^2 + \alpha_2\left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 \\ & + \lambda ||\mathcal{V}||_{*,\text{CD}} + \eta\left\|\mathcal{X} - \mathcal{V} + \frac{\Theta}{2\eta}\right\|_F^2\end{aligned} \tag{21}$$

where $\eta > 0$ is the penalty parameter and Θ is the Lagrangian multiplier. Following ADMM framework, variables $\mathcal{X}$, $\mathcal{V}$, and Θ in (21) can be independently optimized by several tractable subproblems. More specifically,

1) Updating $\mathcal{X}$: The $\mathcal{X}$-subproblem is

$$\begin{aligned}\min_{\mathcal{X}} \ & \alpha_1\left\|\mathbf{Y}_{(3)} - \mathbf{X}_{(3)}\mathbf{BS}\right\|_F^2 + \alpha_2\left\|\mathbf{Z}_{(3)} - \mathbf{R}\mathbf{X}_{(3)}\right\|_F^2 \\ & + \eta\left\|\mathcal{X} - \mathcal{V} + \frac{\Theta}{2\eta}\right\|_F^2.\end{aligned} \tag{22}$$

The objective function in (22) is strongly convex, thus allowing a unique solution. By forcing the derivative of the objective function w.r.t. $\mathcal{X}$ to 0, we can obtain the Sylvester equation

$$\begin{aligned}& \alpha_2\mathbf{R}^{\text{T}}\mathbf{R}\mathbf{X}_{(3)} + \eta\mathbf{X}_{(3)} + \alpha_1\mathbf{X}_{(3)}(\mathbf{BS})(\mathbf{BS})^{\text{T}} \\ & \quad = \alpha_2\mathbf{R}^{\text{T}}\mathbf{Z}_{(3)} + \eta\left(\mathbf{V}_{(3)} - \frac{\mathbf{\Theta}_{(3)}}{2\eta}\right) + \alpha_1\mathbf{Y}_{(3)}(\mathbf{BS})^{\text{T}}\end{aligned} \tag{23}$$

which can be further reformated to the following form:

$$\mathbf{H}_1\mathbf{X}_{(3)} + \mathbf{X}_{(3)}\mathbf{H}_2 = \mathbf{H}_3 \tag{24}$$

where

$$\begin{aligned}\mathbf{H}_1 &= \frac{\alpha_2}{\alpha_1}\mathbf{R}^{\text{T}}\mathbf{R} + \frac{\eta}{\alpha_1}\mathbf{I} \\ \mathbf{H}_2 &= \mathbf{BSS}^{\text{T}}\mathbf{B}^{\text{T}} \\ \mathbf{H}_3 &= \frac{\alpha_2}{\alpha_1}\mathbf{R}^{\text{T}}\mathbf{Z}_{(3)} + \frac{\eta}{\alpha_1}\left(\mathbf{V}_{(3)} - \frac{\mathbf{\Theta}_{(3)}}{2\eta}\right) + \mathbf{Y}_{(3)}(\mathbf{BS})^{\text{T}}.\end{aligned} \tag{25}$$

Since $\mathbf{B} \in \mathbb{R}^{HW\times HW}$ is a block circulant matrix with circulant blocks, $\mathbf{B}$ can be decomposed as follows:

$$\mathbf{B} = \mathbf{FKF}^{-1} \tag{26}$$

where $\mathbf{F}$ and $\mathbf{F}^{-1}$ indicate the fast Fourier transform (FFT) and its inverse FFT (IFFT) matrices, respectively. The diagonal matrix $\mathbf{K} \in \mathbb{C}^{HW\times HW}$ holds the eigenvalues of $\mathbf{B}$ and can be written as follows:

$$\mathbf{K} = \begin{bmatrix}\mathbf{K}_1 & & & \\ & \mathbf{K}_2 & & \\ & & \ddots & \\ & & & \mathbf{K}_r\end{bmatrix} \tag{27}$$

where $\mathbf{K}_i \in \mathbb{C}^{hw\times hw}$, $i = 1, 2, \ldots, r$, and r is the scaling factor. Moreover, $\mathbf{S} \in \mathbb{R}^{HW\times hw}$ satisfies the property $\mathbf{S}^{\text{T}}\mathbf{S} = \mathbf{I}$, and $\mathbf{SS}^{\text{T}}$ is symmetric and idempotent. By leveraging the properties of $\mathbf{B}$ and $\mathbf{S}$, the Sylvester equation (24) can be analytically solved without iteration [31], [41]. The whole

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Closed-Form Solution of (24) for $\mathcal{X}$

Input: $\mathbf{H}_1$, $\mathbf{H}_2$, $\mathbf{H}_3$, $\mathbf{B}$, $\mathbf{S}$.

1: Eigen-decomposition of $\mathbf{B}$: $\mathbf{B} = \mathbf{F}\mathbf{K}\mathbf{F}^{-1}$.
2: $\tilde{\mathbf{K}} = \mathbf{K}(\mathbf{1}_r \otimes \mathbf{1}_{hw})$.
3: Eigen-decomposition of $\mathbf{H}_1$: $\mathbf{H}_1 = \mathbf{Q}_1\mathbf{\Lambda}\mathbf{Q}_1^{-1}$.
4: $\tilde{\mathbf{H}}_3 = \mathbf{Q}_1^{-1}\mathbf{H}_3\mathbf{F}$.
5: **for** $i = 1$ to S **do**
6: $\quad \mathbf{o}_i = \lambda_i^{-1}(\tilde{\mathbf{H}}_3)_i - \lambda_i^{-1}(\tilde{\mathbf{H}}_3)_i\tilde{\mathbf{K}}(\lambda_i r\mathbf{I} + \sum_{j=1}^{r}\mathbf{K}_j^2)\tilde{\mathbf{K}}^{\mathrm{T}}$.
7: **end for**
8: $\mathbf{X}_{(3)} = \mathbf{Q}_1\mathbf{O}\mathbf{F}^{-1}$.

Output: $\mathcal{X}$.

Algorithm 3 ADMM-Based Solver of the CroDoSR Model (18)

Input: LR-HSI $\mathcal{Y}$, HR-MSI $\mathcal{Z}$, $\mathbf{B}$, $\mathbf{S}$, $\mathbf{R}$, C, r, α_1, α_2, λ, and η.

Output: The recovered HR-HSI $\mathcal{X}$.

1: **while** *not converged* **do**
2: Update $\mathcal{X}$ via Algorithm 2;
3: Update $\mathcal{V}$ via (31);
4: Update Θ via (32).
5: **end while**

solving algorithm is summarized in Algorithm 2. Notably, the unitary matrix $\mathbf{Q}_1$ and diagonal matrix $\mathbf{\Lambda}$ are obtained by the eigendecomposition, i.e., $\mathbf{H}_1 = \mathbf{Q}_1\mathbf{\Lambda}\mathbf{Q}_1^{-1}$, where $\mathbf{\Lambda}$ is represented as follows:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_S \end{bmatrix} \tag{28}$$

and $\mathbf{Q}_1$ holds the column eigenvectors of $\mathbf{H}_1$. In addition, $\mathbf{o}_i$ represents the ith row of $\mathbf{O} = \mathbf{Q}_1^{-1}\mathbf{X}_{(3)}\mathbf{F}$, i.e., $\mathbf{O} = [\mathbf{o}_1^{\mathrm{T}}, \mathbf{o}_2^{\mathrm{T}}, \ldots, \mathbf{o}_S^{\mathrm{T}}]^{\mathrm{T}}$, and $\mathbf{1}_r \in \mathbb{R}^r$ is an all-ones vector.

2) Updating $\mathcal{V}$: The $\mathcal{V}$-subproblem is

$$\min_{\mathcal{V}} \ \lambda||\mathcal{V}||_{*,\mathrm{CD}} + \eta\left|\left|\mathcal{X} - \mathcal{V} + \frac{\Theta}{2\eta}\right|\right|_F^2 \tag{29}$$

which is equivalent to

$$\min_{\mathcal{V}} \ \frac{\lambda}{2\eta}||\mathcal{V}||_{*,\mathrm{CD}} + \frac{1}{2}\left|\left|\mathcal{V} - \left(\mathcal{X} + \frac{\Theta}{2\eta}\right)\right|\right|_F^2. \tag{30}$$

Relying upon Theorem 2, the minimizer of $\mathcal{V}$ can be given as follows:

$$\mathcal{V} \leftarrow \mathrm{Prox}_{C,||\cdot||_{*,\mathrm{CD}}}^{\frac{\lambda}{2\eta}}\left(\mathcal{X} + \frac{\Theta}{2\eta}\right). \tag{31}$$

3) Updating Θ: Following to ADMM framework, the Lagrangian multiplier Θ is updated by

$$\Theta \leftarrow \Theta + 2\eta(\mathcal{X} - \mathcal{V}). \tag{32}$$

The overall ADMM-based optimizer of solving the proposed CroDoSR model is summarized in Algorithm 3.

TABLE I
DETAILED CHARACTERIZATION OF ALL COMPARED METHODS

Method	Category	TF type	Is CD?	Transform
CNMF [9]	MF	–	–	–
CSTF [23]	TF	Tucker	–	–
CNN-Fus [15]	MF+DL	–	–	–
TNN [39]	TF	t-SVD	✗	Fourier
F-TNN [33]	TF	t-SVD	✗	framelet
VMTQN [1]	TF	t-SVD	✗	Q
CroDoSR	TF	t-SVD	✓	SLT: Q+framelet

C. Computational Complexity Analysis

In Algorithm 3, the computational burden is mainly related to updating $\mathcal{X}$ and $\mathcal{V}$ subproblems. More specifically, the complexity of updating $\mathcal{X}$ is $\mathcal{O}(HW\max\{\log(HW), S^2\})$. Assume that the SLT C is from Definition 1 with $J_0 = S$, then the $\mathcal{V}$ subproblem requires a complexity of $\mathcal{O}(S\sum_{i=1}^{N-1} J_i J_{i+1} + J_N HWS + J_N\min\{H^2W, W^2H\})$. Accordingly, the total complexity of Algorithm 3 is $\mathcal{O}(HW\max\{\log(HW), S^2\}) + \mathcal{O}(S\sum_{i=1}^{N-1} J_i J_{i+1} + J_N HWS + J_N\min\{H^2W, W^2H\})$. Assume that $N \leq 3$ and $H \approx W \geq S \approx J_i$, $i = 1, 2, \ldots, N$, the above complexity can be further simplified as $\mathcal{O}(J_N H^2 W)$, which grows only subquadratically with the spatial volume HW, thus endowing Algorithm 3 with practical applicability for larger scale data.

VI. Numerical Experiments

In this section, we perform substantial HSI SR experiments to validate the validity and superiority of the CroDoSR model. Mainly, three linear transforms, i.e., Fourier transform [32], framelet transform [33], and Q transform [1], are considered to generate the SLT. In the framelet transform, an additional hyperparameter, i.e., the decomposition level L, is involved. Please see Section VI-D for a detailed parameter analysis. For simplicity, the CroDoSR method with a specified SLT (i.e., Q + framelet) is formed for comparison. More experimental results regarding various SLT combinations are presented in Section VI-D. All numerical experiments are implemented in MATLAB (R2019a) on a computer with 16-Gb RAM and an Intel[3] Core[4] i7-5960X CPU: at 3.00 GHz.

The proposed CroDoSR method (SLT: Q + framelet) is compared with several closely related competitors. In great detail, CNMF [9], the CSTF [23], CNN-Fus [15], TNN [39], F-TNN [33], and VMTQN [1] methods are considered. Please see Table I for detailed information for all comparators. Notably, the TNN, F-TNN, and VMTQN were previously designed for other tasks, and here, we simply employ their corresponding rank constraints for HSI SR, aiming to illustrate the superiority of CD over existing single domains. For a more fair comparison, the involved hyperparameters are fine-tuned or optimally arranged within specified ranges.

To evaluate the performance of simulated experiments, five commonly used metrics, i.e., the peak-signal-to-noise ratio (PSNR, ↑), the structure similarity (SSIM, ↑) [42], the

[3]Registered trademark.
[4]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUALITY METRICS OF ALL COMPARED METHODS ON PAU DATASET UNDER ALL SNRS. (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Method	SNR = 15dB					Method	SNR = 25dB				
	PSNR↑	SSIM↑	UIQI↑	SAM↓	ERGAS↓		PSNR↑	SSIM↑	UIQI↑	SAM↓	ERGAS↓
CNMF [9]	27.87	0.924	0.944	5.11	6.46	CNMF [9]	33.17	0.953	0.964	3.37	3.46
CSTF [23]	28.46	0.946	0.967	4.31	5.97	CSTF [23]	36.97	0.977	0.989	2.51	2.21
CNN-Fus [15]	**28.53**	**0.952**	**0.970**	**4.22**	**5.91**	CNN-Fus [15]	37.43	0.985	**0.993**	2.29	2.10
TNN [39]	27.93	0.915	0.945	5.41	6.28	TNN [39]	34.88	0.945	0.967	4.27	2.99
F-TNN [33]	26.20	0.787	0.870	8.32	7.78	F-TNN [33]	34.93	0.944	0.965	4.27	2.95
VMTQN [1]	28.10	0.917	0.953	5.23	6.11	VMTQN [1]	35.10	0.947	0.969	4.15	2.83
CroDoSR	28.50	0.949	**0.970**	4.31	5.92	CroDoSR	**37.72**	**0.986**	**0.993**	**2.17**	**2.04**
Method	SNR = 35dB					Method	SNR = 45dB				
	PSNR↑	SSIM↑	UIQI↑	SAM↓	ERGAS↓		PSNR↑	SSIM↑	UIQI↑	SAM↓	ERGAS↓
CNMF [9]	34.97	0.961	0.970	3.03	2.85	CNMF [9]	35.28	0.961	0.970	2.87	2.68
CSTF [23]	41.07	0.982	0.992	2.19	1.37	CSTF [23]	41.75	0.982	0.992	2.19	1.28
CNN-Fus [15]	42.65	0.989	0.994	1.84	1.16	CNN-Fus [15]	43.80	0.988	0.994	1.88	1.06
TNN [39]	37.94	0.949	0.969	4.13	2.42	TNN [39]	38.58	0.950	0.970	4.15	2.36
F-TNN [33]	38.01	0.949	0.970	4.14	2.38	F-TNN [33]	38.81	0.950	0.970	4.13	2.21
VMTQN [1]	38.23	0.951	0.972	4.02	2.23	VMTQN [1]	39.28	0.952	0.977	3.96	1.86
CroDoSR	**43.56**	**0.991**	**0.996**	**1.71**	**1.06**	CroDoSR	**44.69**	**0.990**	**0.995**	**1.76**	**0.95**

universal quality index (UIQI, ↑) [43], the spectral angle mapper (SAM, ↓), and the relative dimensionless global error in synthesis (ERGAS, ↓), are adopted.

A. Simulated Datasets

The performance is comprehensively validated on four publicly available datasets, i.e., *Pavia University* (PaU[5]), *Pavia City Center* (PaC[6]), CAVE,[7] and Harvard.[8]

The PaU dataset is sized 610 × 340 × 115. By removing the bands with a low signal-to-noise ratio (SNR), only 93 bands are preserved. Subsequently, the ground-truth (GT) data is cropped from the up-left 320 × 320 image cube, i.e., $H = 320$, $W = 320$, and $S = 93$. The LR-HSI $\mathcal{Y}$ is simulated by Gaussian blurring with 7 × 7 kernel (standard deviation: 2) and downsampling with $r = 4$. Meanwhile, the HR-MSI $\mathcal{Z}$ is generated by applying the IKONOS-like reflectance spectral response to the GT image.

The PaC dataset consists of both 610 × 340 spatial pixels and 93 bands. For time-saving validation, we resize the HSI of PaC to 200 × 200 × 80 in the experiments. Afterward, the LR-HSI and HR-MSI are similarly simulated as in the PaU case.

The CAVE dataset includes 31 HSIs with a size of 512 × 512 × 31. In numerical experiments, we empirically select five HSIs (i.e., *Flower*, *Clay*, *Cloth*, *CD*, and *Superball*) from the CAVE dataset as the GT images. To obtain the simulated LR-HSIs, we process the GT images by applying a 7 × 7 Gaussian kernel (standard deviation: 2) and downsampling on each $r \times r$ pixels, where r is configured as 8, 16, and 32. In addition, the HR-MSIs are simulated by degenerating the GT images in spectral using the spectral response of a Nikon D700 camera.

The Harvard dataset includes 50 HSIs, each HSI has a spatial resolution of 1392 × 1040 and 31 spectral bands. Also, we empirically select five HSIs, then we crop the up-left 512 × 512 cubes as the GT images, i.e., $H = 512$, $W = 512$, and $S = 31$. Similar to the CAVE case, the LR-HSIs and HR-MSIs are generated.

Remark 2: To model a more realistic experimental condition, Gaussian noise with the same SNR is simultaneously added to the simulated LR-HSIs and HR-MSIs. For the PaU and PaC datasets, the SNR is varied from 15 to 45 dB with an interval of 10 dB. Instead, the SNR for the CAVE and Harvard datasets is uniquely designated as 45 dB.

B. Numerical Results on Simulated Datasets

1) Results on PaU Dataset: The visual comparison of the PaU dataset with SNR = 45 dB is presented in Fig. 3. As shown in the enlarged areas of Fig. 3, the CroDoSR preserves more details compared with other methods, thus yielding better recovery performance. From Fig. 4(a), we also observe that the CroDoSR holds the highest PSNR values for almost all bands, showing its numerical superiority. To further measure the spectral quality, we show the spectral curves of the [186, 1]th pixel for the images obtained by the seven methods in Fig. 8(a). It can be seen that the spectral curve of the CroDoSR is closer to the original spectral curve than other compared methods in most bands. Based on the above analysis, the CroDoSR method can well preserve both spatial and spectral information. Quantitatively, Table II presents the metrics of different methods on PaU dataset under all SNRs. From Table II, we can observe that single-domain-based methods, i.e., the TNN, F-TNN, and VMTQN, generate an inferior performance. Since the CNN-Fus employs a DL denoiser, it obtains better results than other compared methods on low SNR cases. Comparatively, the CroDoSR achieves competitive results for lower SNRs and outperforms all compared methods for higher SNRs.

2) Results on PaC Dataset: Similar to the PaU experiment, Figs. 4(b), 5, and 8(b) provide the SR results of different methods on PaC dataset with SNR = 45 dB. According to the related results, the CroDoSR achieves finer visualization,

[5]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes#Pavia_Centre_and_University
[6]http://www.ehu.eus/ccwintco/index.php
[7]https://www.cs.columbia.edu/CAVE/databases/multispectral/
[8]https://oir.harvard.edu/common-data-set

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

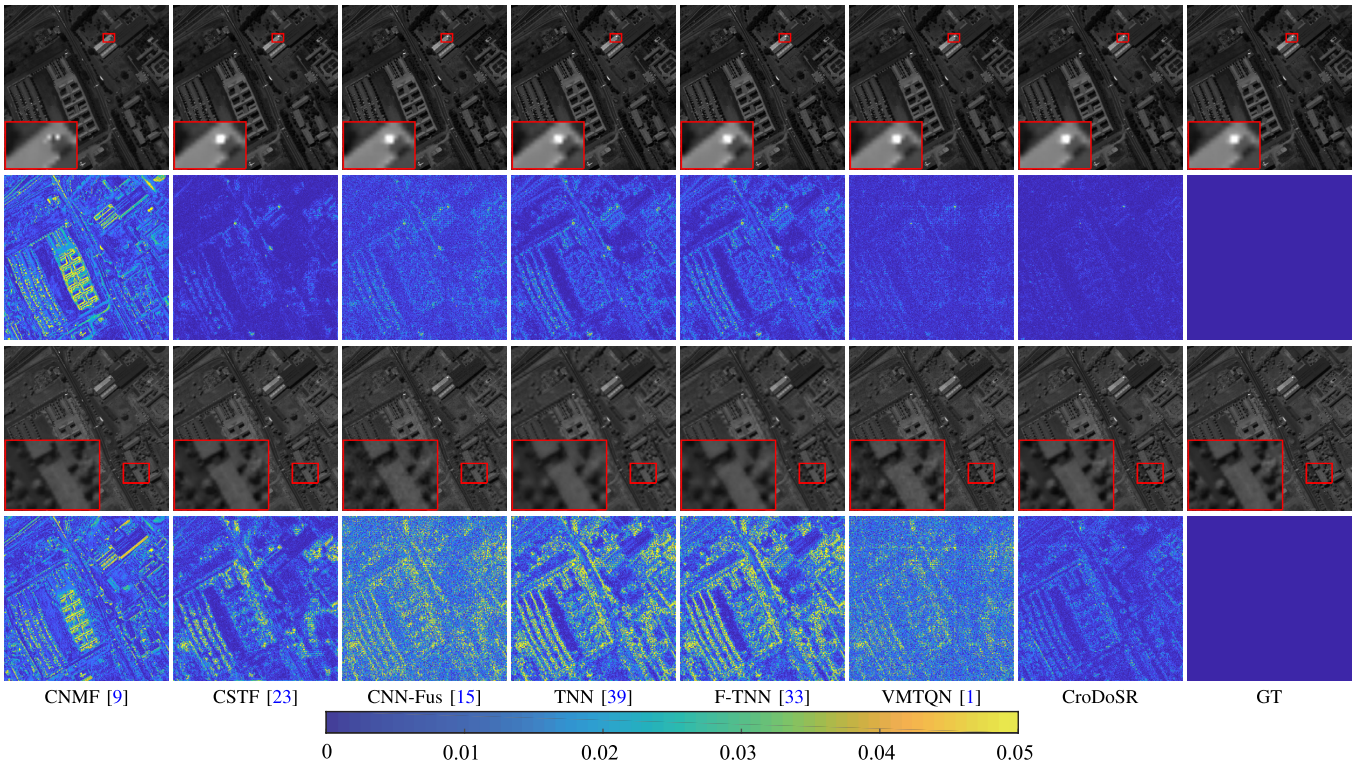


Fig. 3. Visual performance of all competitors on PaU dataset when SNR = 45 dB. The first two rows are the reconstructed images and the corresponding error maps for the 58th spectral band. The last two rows are the reconstructed images and the corresponding error maps for the 65th spectral band.

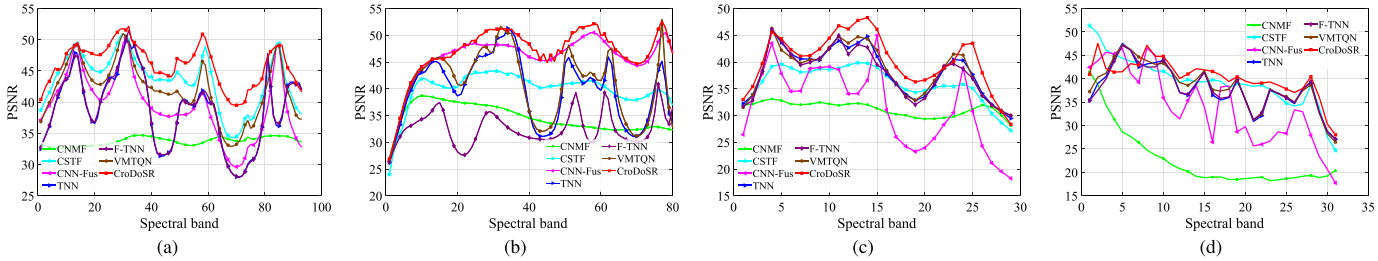


Fig. 4. PSNR curves of all recovered images on (a) PaU dataset (SNR = 45 dB), (b) PaC dataset (SNR = 45 dB), (c) *Flower* (source: CAVE, $r = 32$), and (d) *Imgb4* (source: Harvard, $r = 32$), respectively.

competitive metrics, and more realistic spectral structure compared with other methods, indicating its superiorities for better preserving spatial and spectral details. Furthermore, Fig. 6 shows the PSNR, SSIM, UIQI, SAM, and ERGAS values of different methods on PaC dataset under all SNRs. For any verified case, the CroDoSR always achieves the best performance in terms of these quantitative metrics.

3) Results on CAVE and Harvard Datasets: For different scaling factors (i.e., $r = 8$, 16, and 32), we first evaluate the numerical performance across five experiments on the CAVE and Harvard datasets, as tabulated in Table III. From Table III, we observe that the CroDoSR method typically achieves the optimal metrics. Compared with the TNN, F-TNN, and VMTQN, the CroDoSR achieves higher PSNR and SSIM values, indicating the higher similarity of the acquired image by the CroDoSR with the GT image. Especially, the superior SAM and ERGAS metrics directly indicate the better spectral and spatial preservation of the proposed CroDoSR method, respectively.

Regarding the qualitative comparisons, the visual inspections of these methods on CAVE and Harvard datasets are displayed in Figs. 7 and 9 when $r = 32$, respectively. Apparently, the CroDoSR realizes more image details, such as edges, than other methods. Also, the fewer residuals in error maps verify the better visual performance of the CroDoSR. Moreover, Fig. 4(c) and (d) shows the PSNR curves of the obtained images on CAVE and Harvard datasets, respectively. According to Fig. 4(c) and (d), the CroDoSR still achieves the best PSNR performances, demonstrating its outstanding capability. Likewise, we investigate the spectral curves of all compared methods on CAVE and Harvard datasets, as demonstrated in Fig. 8(c) and (d). From Fig. 8(c) and (d), the spectral curves of the CroDoSR are closer to the GT ones, which confirms each other with the SAM metrics.

The above experiments comprehensively justify that the CroDoSR can realize the superior SR performance, both spatially and spectrally. In particular, the CroDoSR is extremely robust to various datasets, SNR intensities, and scaling factors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

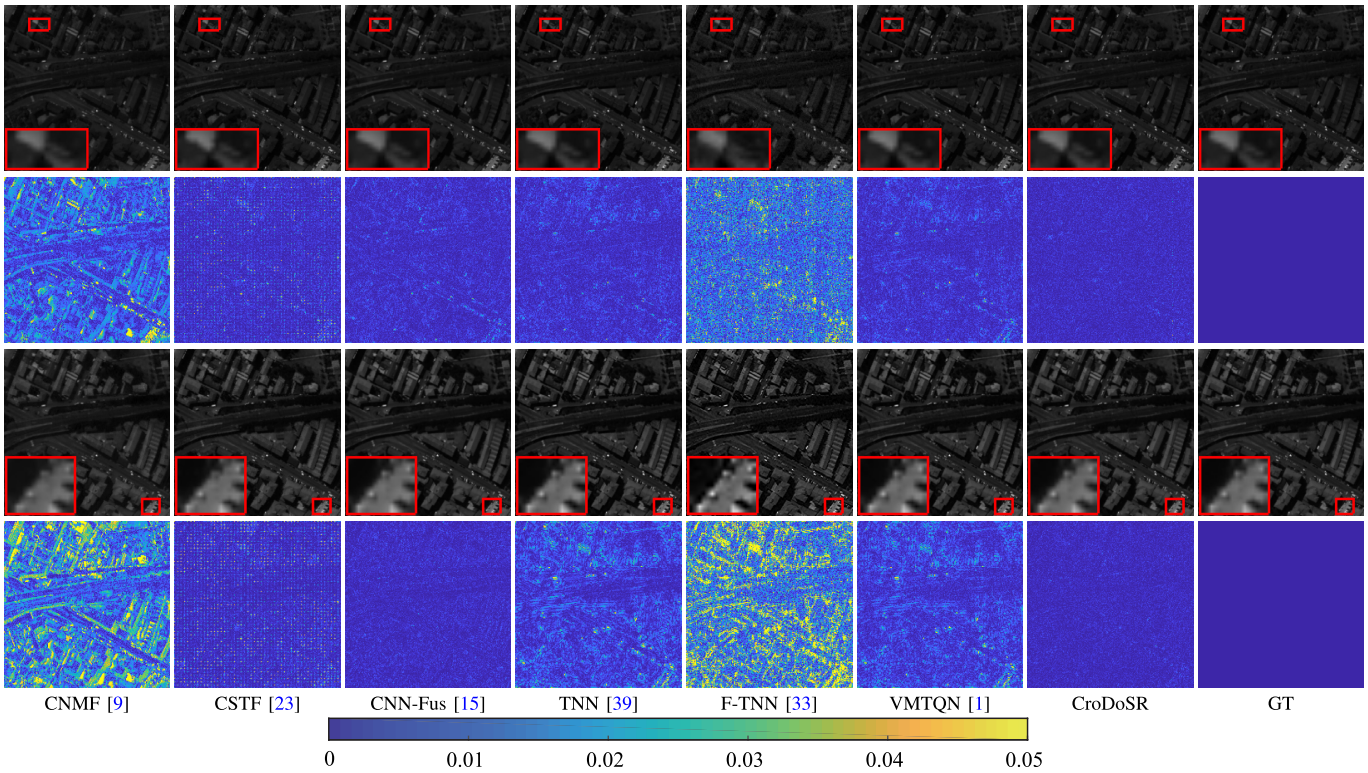


Fig. 5. Visual performance of all competitors on PaC dataset when SNR = 45 dB. The first two rows are the reconstructed images and the corresponding error maps for the 30th spectral band. The last two rows are the reconstructed images and the corresponding error maps for the 58th spectral band.

TABLE III

QUALITY METRICS OF ALL COMPARED METHODS ON CAVE AND HARVARD DATASETS USING VARIOUS SCALING FACTORS. EACH METRIC VALUE IS THE AVERAGE OF FIVE INDEPENDENT SR EXPERIMENTS (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Dataset	r	Metric	Pixel-domain			Single-domain			Cross-domain
			CNMF [9]	CSTF [23]	CNN-Fus [15]	TNN [39]	F-TNN [33]	VMTQN [1]	CroDoSR
CAVE	8	PSNR↑	36.04±2.93	39.98±3.28	36.70±3.28	42.04±3.28	42.11±3.28	42.34±3.28	**43.84±3.28**
		SSIM↑	0.956±0.021	0.947±0.012	0.929±0.050	0.977±0.008	0.980±0.009	0.980±0.008	**0.983±0.005**
		SAM↓	9.29±3.35	13.10±5.14	9.89±2.03	9.51±4.16	8.02±3.72	8.41±3.29	**7.84±3.38**
		ERGAS↓	2.38±0.69	0.89±0.39	1.41±0.35	0.88±0.45	0.89±0.41	0.86±0.39	**0.81±0.46**
	16	PSNR↑	29.01±1.89	36.90±3.48	32.92±3.81	38.48±2.83	37.71±3.91	37.81±3.93	**39.00±4.23**
		SSIM↑	0.892±0.038	0.928±0.026	0.920±0.030	0.964±0.008	0.967±0.007	0.966±0.007	**0.969±0.004**
		SAM↓	12.49±3.81	17.07±7.36	17.63±2.11	11.30±5.17	**9.46±3.59**	10.12±4.16	10.56±4.61
		ERGAS↓	2.91±0.40	1.43±0.67	2.56±0.73	1.24±0.63	1.26±0.63	1.25±0.63	**1.11±0.64**
	32	PSNR↑	25.87±2.63	36.19±3.28	32.02±3.74	35.93±3.25	35.94±3.78	35.99±3.80	**36.68±4.07**
		SSIM↑	0.804±0.018	0.929±0.015	0.922±0.023	0.947±0.009	0.951±0.008	0.951±0.007	**0.952±0.007**
		SAM↓	17.26±4.01	18.38±6.77	20.46±2.66	13.69±5.17	12.65±4.36	12.88±4.55	**12.55±4.99**
		ERGAS↓	2.90±0.56	1.97±0.56	3.30±1.07	1.62±0.50	1.60±0.51	1.56±0.53	**1.33±0.54**
Harvard	8	PSNR↑	29.73±4.04	34.81±6.05	34.09±5.88	35.34±5.18	35.37±5.90	36.25±6.37	**36.45±6.42**
		SSIM↑	0.838±0.049	0.875±0.080	0.897±0.042	0.925±0.044	0.925±0.045	0.916±0.057	**0.928±0.048**
		SAM↓	6.08±1.82	8.37±3.29	7.22±1.84	5.42±1.50	5.43±1.54	6.13±2.12	**5.04±1.61**
		ERGAS↓	1.68±0.29	0.60±0.34	0.72±0.22	0.82±0.22	0.83±0.22	0.70±0.21	**0.55±0.25**
	16	PSNR↑	26.40±2.76	34.25±6.28	32.40±5.66	32.40±4.42	34.03±5.67	35.20±6.25	**35.53±5.97**
		SSIM↑	0.789±0.056	0.875±0.081	0.899±0.042	0.915±0.026	0.916±0.048	0.913±0.057	**0.924±0.049**
		SAM↓	6.58±1.58	9.56±3.73	10.82±3.84	6.65±1.62	6.63±1.64	6.86±2.00	**5.65±1.65**
		ERGAS↓	2.56±0.63	1.18±0.59	1.42±0.39	1.43±0.47	1.49±0.44	1.10±0.49	**1.02±0.46**
	32	PSNR↑	23.20±2.65	34.09±6.19	32.05±5.67	32.05±4.42	33.18±5.26	33.68±5.47	**35.27±6.15**
		SSIM↑	0.701±0.080	0.886±0.072	0.904±0.044	0.910±0.051	0.911±0.050	0.901±0.061	**0.919±0.053**
		SAM↓	7.87±1.57	9.50±3.72	12.46±4.44	7.82±1.86	7.80±1.88	8.04±2.13	**6.61±1.93**
		ERGAS↓	3.28±1.24	2.14±1.18	2.28±0.92	2.40±0.91	2.42±0.92	**1.85±1.04**	2.01±0.94

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

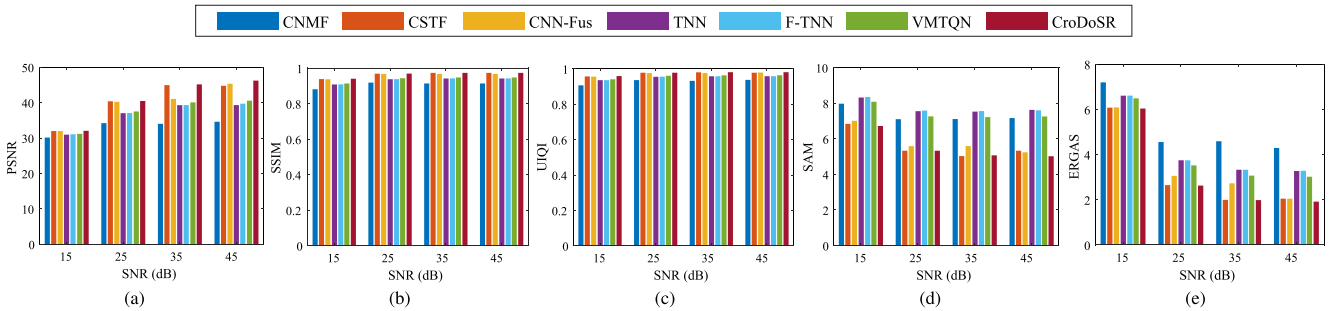


Fig. 6. Statistical values of all compared methods for (a) PSNR, (b) SSIM, (c) UIQI, (d) SAM, and (e) ERGAS on PaC dataset under all SNRs.

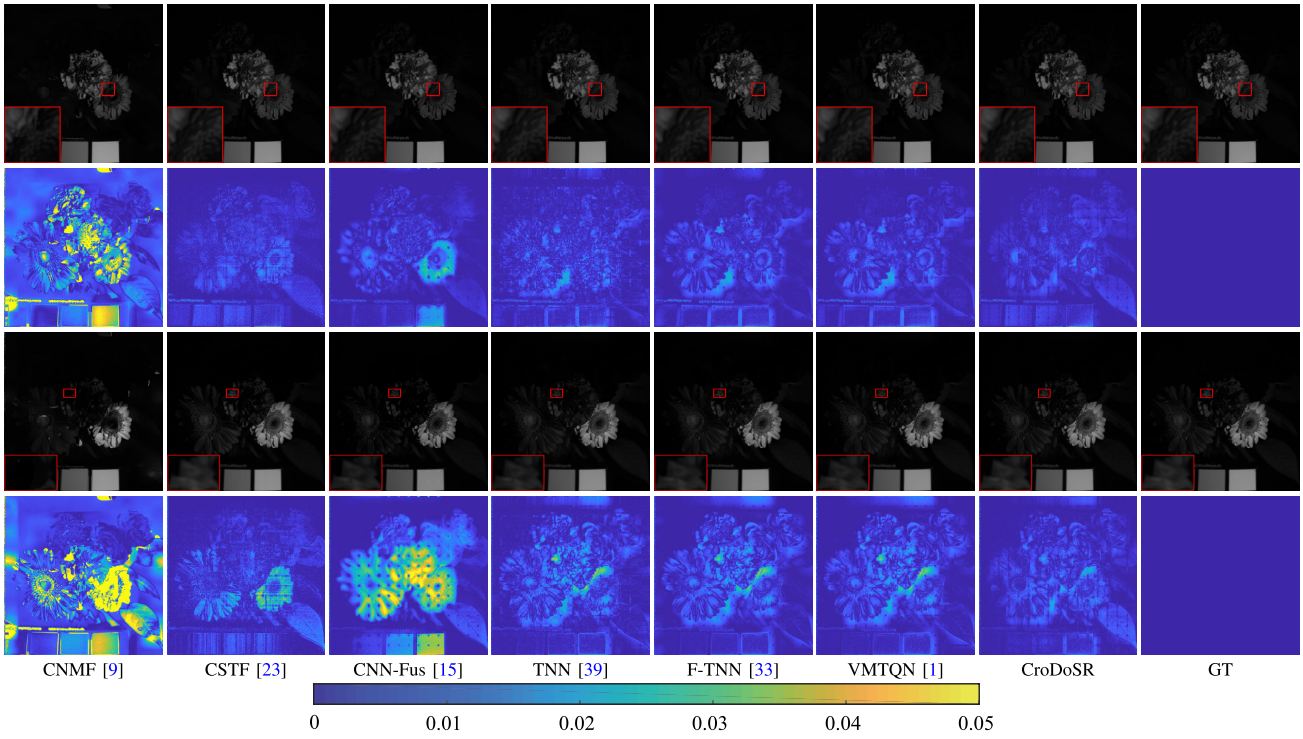


Fig. 7. Visual performance of all competitors on *Flower* (source: CAVE dataset) when $r = 32$. The first two rows are the reconstructed images and the related error maps for the fourth spectral band. The last two rows are the reconstructed images and the related error maps for the 13th spectral band.

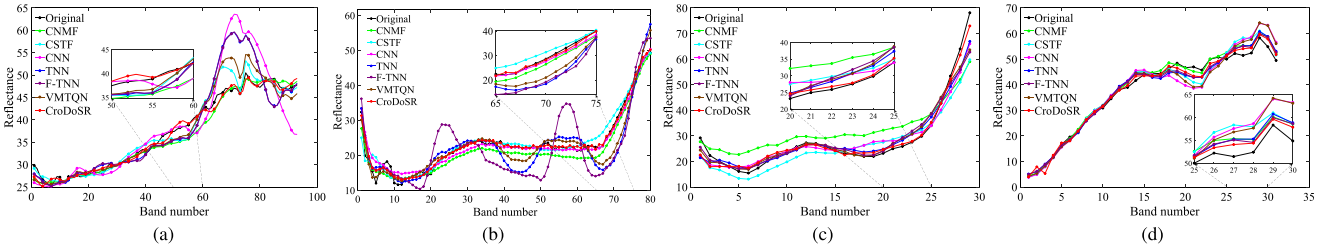


Fig. 8. Spectral curves of all recovered images for [186, 1]th spatial pixel on (a) PaU dataset (SNR = 45 dB), (b) PaC dataset (SNR = 45 dB), (c) *Flower* (source: CAVE, $r = 32$), and (d) *Imgb4* (source: Harvard, $r = 32$), respectively.

C. Numerical Results on Real Dataset

Regarding the real HSI SR experiment, the DFC2018 Houston dataset,[9] which is released by the 2018 IEEE GRSS Data Fusion Contest [44], is adopted. The Houston dataset contains both an LR-HSI and an HR-MSI (i.e., an RGB image). The LR-HSI acquired from an ITRES CASI-1500 sensor is of size 4172 × 1202 × 48 with a 1 m spatial resolution and 380–1050 nm spectral coverage. The HR-MSI acquired from a DiMAC ULTRALIGHT+ sensor is of size 83 440 × 24 040 × 3 with a 0.05 m spatial resolution. In our experiment, the downsampled images of the edge-lightly-cropped LR-HSI and HR-MSI with the sizes

[9]https://hyperspectral.ee.uh.edu/?page_id=1075

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

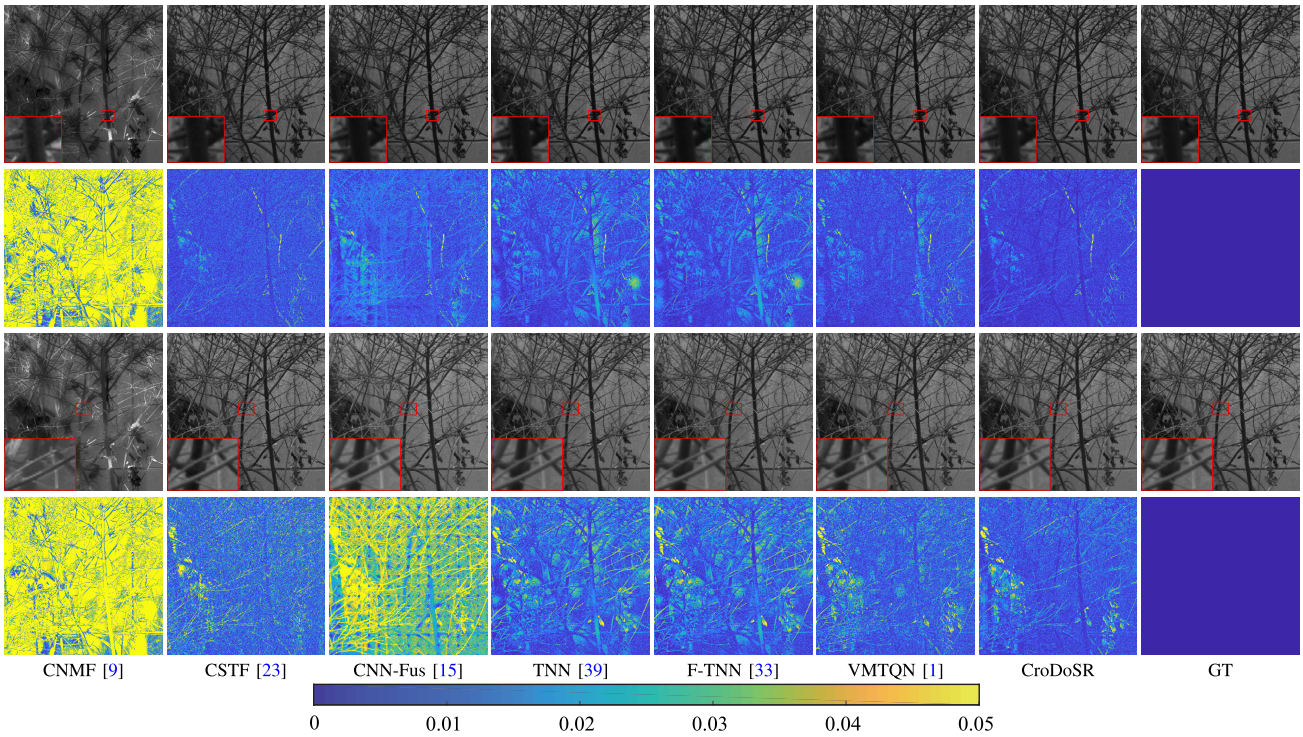


Fig. 9. Visual performance of all competitors on *Imgb4* (source: Harvard dataset) when $r = 32$. The first two rows are the reconstructed images and the related error maps for the 14th spectral band. The last two rows are the reconstructed images and the related error maps for the 25th spectral band.

TABLE IV

QUALITY METRICS OF THE CRODOSR MODELS WITH/WITHOUT CD-RANK MINIMIZATION ON BOTH PAU AND PAC DATASETS WHEN SNR = 45 dB (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Method	PaU dataset			PaC dataset		
	PSNR↑	SAM↓	ERGAS↓	PSNR↑	SAM↓	ERGAS↓
CroDoSR ($\lambda = 0$)	38.742	4.062	2.355	39.853	7.500	3.294
CroDoSR ($\lambda > 0$)	**44.694**	**1.758**	**0.946**	**46.220**	**5.016**	**1.921**

of $695 \times 200 \times 48$ and $2780 \times 800 \times 3$ are adopted for validation. That is, the scaling factor $r = 4$. In this case, since the blurring matrix and spectral response are not directly available, we alternately employ the method proposed in [45] to estimate their approximations. Moreover, the hybrid quality with no reference (HQNR, ↑) [46], which consists of both the spectral distortion index D_λ and the spatial distortion index D_s [47], is used for quantitative evaluation. For D_s, the value is computed with the processed gray-scaled image obtained by merging the RGB spectral channels of the HR-MSI.

The visual results and metrics of all the compared methods are presented in Fig. 10. From the reconstructed HSI images, we observe that the CNMF achieves over-emphasized details and distorted spectral characteristics, and the CSTF and CNN-Fus are both visually inferior. In contrast, the TNN, F-TNN, VMTQN, and CroDoSR realize more desirable performances. From the metrics and the corresponding HQNR maps, the proposed CroDoSR further demonstrates the incremental advantages of spatial characterization with spectral preservation, proving its effectiveness in real HSI SR applications.

TABLE V

METRIC COMPARISON ON PAU DATASET (SNR = 25 AND 45 dB) BY USING AN AVERAGE BLURRING KERNEL OF SIZE 5×5 (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Method	SNR = 25dB				
	PSNR↑	SSIM↑	UIQI↑	SAM↓	ERGAS↓
CNMF [9]	32.50	0.953	0.963	3.71	3.63
CSTF [23]	36.65	0.972	0.987	2.78	2.34
CNN-Fus [15]	37.46	0.984	**0.990**	2.27	2.09
TNN [39]	33.67	0.926	0.945	5.09	3.56
F-TNN [33]	33.13	0.901	0.936	5.42	3.68
VMTQN [1]	34.30	0.930	0.960	4.78	3.08
CroDoSR	**37.64**	**0.985**	**0.990**	**2.22**	**2.05**
Method	SNR = 45dB				
	PSNR↑	SSIM↑	UIQI↑	SAM↓	ERGAS↓
CNMF [9]	35.45	0.953	0.964	3.69	3.21
CSTF [23]	41.29	0.979	0.990	2.38	1.37
CNN-Fus [15]	44.06	0.990	0.990	1.78	1.00
TNN [39]	36.49	0.931	0.953	5.03	3.04
F-TNN [33]	36.74	0.930	0.959	4.95	2.78
VMTQN [1]	37.34	0.935	0.963	4.73	2.46
CroDoSR	**44.91**	**0.991**	**0.996**	**1.71**	**0.92**

D. Discussion

1) Parameter Analysis: In Algorithm 3, four parameters are actually involved, i.e., the regularization parameters α_1, α_2, λ, and the penalty parameter η. Moreover, since the framelet transform is adopted, the framelet level L also needs to be decided. For our experiments, α_1, α_2, and η are relatively stable, thus we empirically specify $\alpha_1 = \alpha_2 = 1$ and $\eta = 10^{-3}$ in all experiments. Consequently, only λ and L should further

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

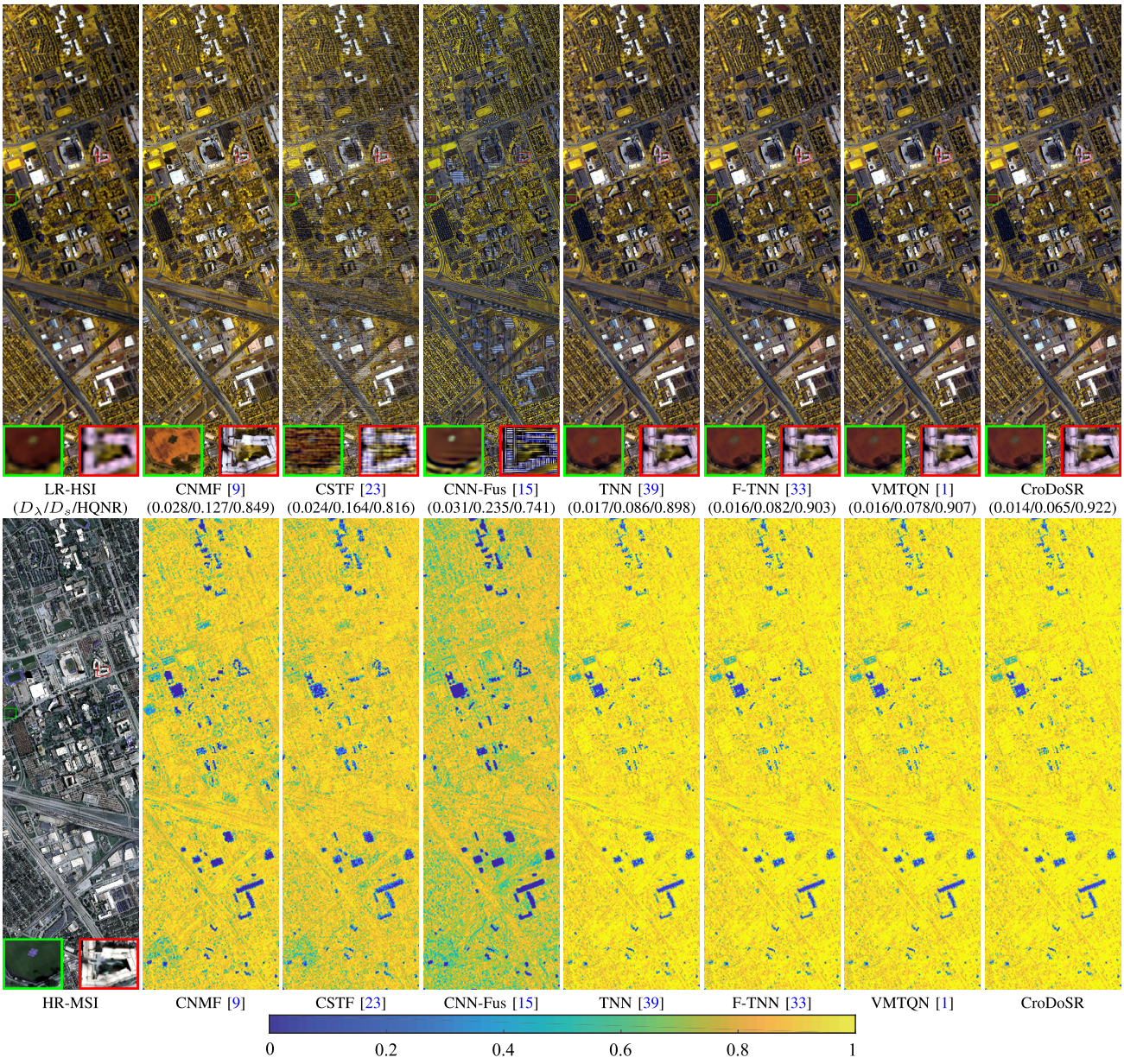


Fig. 10. Visual performance of all competitors on the real DFC2018 Houston dataset. The first column is the available LR-HSI and HR-MSI. The remaining columns are the reconstructed images by different methods and the related HQNR maps. (D_λ/D_s/HQNR) is attached on the top of all HQNR maps. The 48th, 32nd, and 16th spectral bands of HSI images are rendered as R, G, and B channels, respectively.

be considered. Taking the HSI SR of PaC dataset (SNR = 15, 35 dB) as an example, we evaluate the performance of the CroDoSR method with varying λ and L.

From Fig. 11(a) and (b), when λ remains the same, the PSNR is optimal under $L = 1$. Instead of fixing the L, the PSNR is higher for $\lambda < 0.1$. Accordingly, λ is selected from the candidate set $\{0.005, 0.02, 0.1\}$, and L is designated as 1 for all experiments.

2) Runtime Analysis: According to the computational complexity analysis in Section V-C, when the size $H \times W \times S$ is given, the computational burden of the CroDoSR is directly determined by the SLT. Intuitively, the collection $\{L_i\}_{i=1}^N$ with a larger N will inevitably consume more computing time. In Fig. 12, we report the computing time (/s: seconds) on PaC dataset under SNR = 45 dB. As observed, the CroDoSR (SLT: Q + framelet) requires more computing time than others, especially the F-TNN and VMTQN.

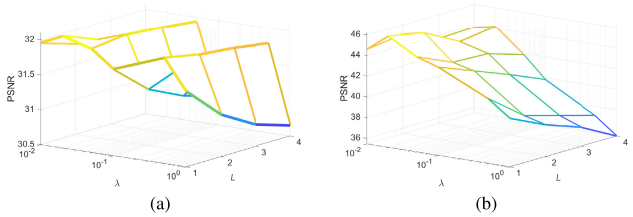


Fig. 11. Influence analysis of the regularization parameter λ and the framelet level L when (a) SNR = 15 dB and (b) SNR = 35 dB.

3) Validity of Tensor CD-Rank Regularization: Let $\lambda = 0$, then the CroDoSR model degenerates to the original HSI SR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
NUMERICAL RESULTS OF THE ABLATION STUDY FOR DIFFERENT SLTS ON CAVE DATASET WHEN $r = 8$ (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Case	$CD_{Fourier\circ framelet}$	$CD_{framelet\circ Fourier}$	$CD_{Fourier\circ Q}$	$CD_{Q\circ Fourier}$	$CD_{Q\circ framelet}$	$CD_{framelet\circ Q}$
PSNR	42.32	42.40	42.52	42.56	43.81	**43.84**
Case	$CD_{Q\circ Fourier\circ framelet}$	$CD_{Q\circ framelet\circ Fourier}$	$CD_{framelet\circ Fourier\circ Q}$	$CD_{framelet\circ Q\circ Fourier}$	$CD_{Fourier\circ Q\circ framelet}$	$CD_{Fourier\circ framelet\circ Q}$
PSNR	42.78	42.96	42.37	42.88	39.07	40.80

TABLE VII
NUMERICAL RESULTS AND COMPUTATIONAL TIME (/h: HOURS) FOR ADDITIONAL COMPARISON ON THE LARGER SIZE PAU DATASET. (**BOLD**: BEST AND UNDERLINE: SECOND BEST)

Method	PSNR↑	SSIM↑	UIQI↑	SAM↓	ERGAS↓	Time/h
FSTRD [48]	43.77	0.986	0.992	2.48	1.35	0.48
NF-3DLogTNN [49]	45.20	**0.991**	**0.994**	**2.01**	**0.94**	16.16
ZSL [17]	**45.43**	0.983	0.984	2.30	1.40	–
CroDoSR	44.57	0.988	**0.994**	2.23	1.08	0.11

model without regularization term. Numerically, Table IV provides the metric values of the CroDoSR models with/without CD-rank constraints on PaU and PaC datasets when SNR = 45 dB. Following Table IV, the validity of tensor CD-rank minimization for HSI SR can be clearly confirmed.

4) Influence of Convolutional Kernel: In numerical experiments, the blurring matrix is simulated by a Gaussian rather than any other kernel. Therefore, we here perform the additional experiments on the PaU dataset (SNR = 25, 45 dB) by using an average blurring kernel of size 5×5. From Table V, we can observe that the CroDoSR can still achieve the best metrics, indicating its robustness for convolutional kernels.

5) Ablation Study for SLT: For a deeper insight into CD, we further investigate the influence of various SLTs. To this end, 12 cases consisting of Fourier [50], framelet [39], and Q [1] are tested on CAVE dataset when $r = 8$, see Table VI. By comparing the results in Tables III and VI, we can experimentally demonstrate the superiority of CD over single domain, while the benefit is more significant if the sequence of transforms is properly specified. For instance, the $CD_{framelet\circ Q}$ (i.e., SLT: Q + framelet) is adopted in the CroDoSR model for better numerical performance.

6) Additional Comparison: For a more comprehensive evaluation, we additionally select two tensor decomposition-based methods (i.e., FSTRD [48] and NF-3DLogTNN [49]) and one DL-based method (i.e., ZSL [17]) for further comparison. A larger size (i.e., $608 \times 336 \times 93$) subimage of the PaU dataset is employed to simulate the LR-HSI and HR-MSI with a 7×7 Gaussian kernel (standard deviation: 3) and downsampling factor $r = 4$. In this comparison, the iteration number of the CroDoSR method is reduced to 20. Interestingly, fewer iterations do not significantly affect the performance (e.g., PSNR loss: 2.74%), but rather effectively improve the computational efficiency, making the CroDoSR more applicable for larger HSI data. The results and computation time (/h: hours) are presented in Table VII. According to the results, although the nonlocal-based NF-3DLogTNN method reflects an incremental advantage over the others, it is extremely computationally resource consuming. Furthermore, when compared with the tensor ring decomposition-based FSTRD and DL-based ZSL methods, the proposed CroDoSR method can still realize comparable and competitive performance with an acceptable computational burden.

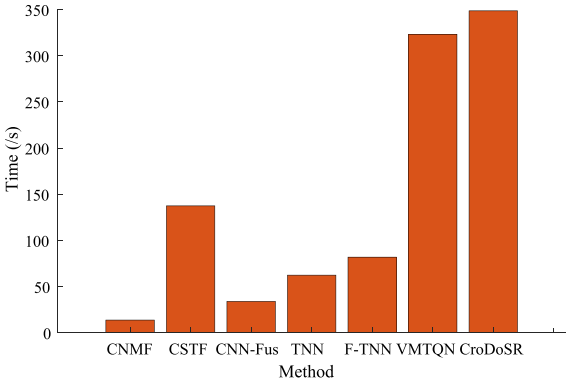


Fig. 12. Computation time (/s: seconds) of all compared methods on PaC dataset under SNR = 45 dB.

VII. CONCLUSION

In this article, we proposed a t-SVD-based method for HSI SR, called CroDoSR. Specifically, we first defined the basic concepts of SLT and CD, then explored the lower-rank potential of the underlying HR-HSI on CD. To characterize such a property, we developed the CD-oriented tensor rank, termed tensor CD rank. Correspondingly, we deduced a tensor CD-nuclear norm as the convex relaxation of the tensor CD rank. By embedding the CD-nuclear norm, we formulated the resulting CD-rank minimization model, i.e., CroDoSR. To optimize the proposed model, we designed an efficient ADMM-based solving algorithm. Experimental results on simulated and real datasets confirmed the effectiveness of the CroDoSR model and also revealed the superiority of CD over single domain for the HSI SR problem.

REFERENCES

[1] H. Kong, C. Lu, and Z. Lin, "Tensor Q-rank: New data dependent definition of tensor rank," *Mach. Learn.*, vol. 110, no. 7, pp. 1867–1900, Jul. 2021.
[2] S. Yang, M. Wang, Z. Feng, Z. Liu, and R. Li, "Deep sparse tensor filtering network for synthetic aperture radar images classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3919–3924, Aug. 2018.
[3] X. Cao, J. Yao, Z. Xu, and D. Meng, "Hyperspectral image classification with convolutional neural network and active learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4604–4616, Jul. 2020.
[4] D. Zhu, B. Du, and L. Zhang, "Learning single spectral abundance for hyperspectral subpixel target detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 1–11, Jan. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

[5] H. Qin, W. Xie, Y. Li, and Q. Du, "HTD-TS^3: Weakly supervised hyperspectral target detection based on transformer via spectral–spatial similarity," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Aug. 2, 2023, doi: 10.1109/TNNLS.2023.3298145.

[6] Z. Pan, G. Healey, M. Prasad, and B. Tromberg, "Face recognition in hyperspectral images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 12, pp. 1552–1560, Dec. 2003.

[7] M. Uzair, A. Mahmood, and A. Mian, "Hyperspectral face recognition with spatiospectral information fusion and PLS regression," *IEEE Trans. Image Process.*, vol. 24, no. 3, pp. 1127–1137, Mar. 2015.

[8] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 2329–2336.

[9] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[10] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3586–3594.

[11] Q. Wei, J. Bioucas-Dias, N. Dobigeon, J. Tourneret, M. Chen, and S. Godsill, "Multiband image fusion based on spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7236–7249, Dec. 2016.

[12] L. Zhang, W. Wei, C. Bai, Y. Gao, and Y. Zhang, "Exploiting clustering manifold structure for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5969–5982, Dec. 2018.

[13] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Hyperspectral and multispectral image fusion via deep two-branches convolutional neural network," *Remote Sens.*, vol. 10, no. 5, p. 800, 2018.

[14] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

[15] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.

[16] J.-F. Hu, T.-Z. Huang, L.-J. Deng, T.-X. Jiang, G. Vivone, and J. Chanussot, "Hyperspectral image super-resolution via deep spatiospectral attention convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7251–7265, Dec. 2022.

[17] R. Dian, A. Guo, and S. Li, "Zero-shot hyperspectral sharpening," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12650–12666, Oct. 2023.

[18] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[19] X. Wang, J. Chen, Q. Wei, and C. Richard, "Hyperspectral image super-resolution via deep prior regularization with parameter estimation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 4, pp. 1708–1723, Apr. 2022.

[20] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522412.

[21] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Apr. 2023, Art. no. 5509417.

[22] H. V. Nguyen, M. O. Ulfarsson, J. R. Sveinsson, and M. D. Mura, "Deep SURE for unsupervised remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412613.

[23] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[24] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5344–5353.

[25] C. Yi, Y.-Q. Zhao, and J. C. Chan, "Hyperspectral image super-resolution based on spatial and spectral correlation fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 7, pp. 4165–4177, Jul. 2018.

[26] R. Dian, S. Li, L. Fang, T. Lu, and J. M. Bioucas-Dias, "Nonlocal sparse tensor factorization for semiblind hyperspectral and multispectral image fusion," *IEEE Trans. Cybern.*, vol. 50, no. 10, pp. 4469–4480, Oct. 2020.

[27] Y. Xu, Z. Wu, J. Chanussot, P. Comon, and Z. Wei, "Nonlocal coupled tensor CP decomposition for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 348–362, Mar. 2019.

[28] K. Wang, Y. Wang, X. Zhao, J. C. Chan, Z. Xu, and D. Meng, "Hyperspectral and multispectral image fusion via nonlocal low-rank tensor decomposition and spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7654–7671, Nov. 2020.

[29] W. He, Y. Chen, N. Yokoya, C. Li, and Q. Zhao, "Hyperspectral super-resolution via coupled tensor ring factorization," *Pattern Recognit.*, vol. 122, Feb. 2022, Art. no. 108280.

[30] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

[31] R. Dian and S. Li, "Hyperspectral image super-resolution via subspace-based low tensor multi-rank regularization," *IEEE Trans. Image Process.*, vol. 28, no. 10, pp. 5135–5146, Oct. 2019.

[32] M. E. Kilmer and C. D. Martin, "Factorization strategies for third-order tensors," *Linear Algebra Appl.*, vol. 435, no. 3, pp. 641–658, Aug. 2011.

[33] T. Jiang, M. K. Ng, X. Zhao, and T. Huang, "Framelet representation of tensor nuclear norm for third-order tensor completion," *IEEE Trans. Image Process.*, vol. 29, pp. 7233–7244, 2020.

[34] Y. Chen, W. He, N. Yokoya, and T.-Z. Huang, "Hyperspectral image restoration using weighted group sparsity-regularized low-rank tensor decomposition," *IEEE Trans. Cybern.*, vol. 50, no. 8, pp. 3556–3570, Aug. 2020.

[35] Y. Chen, W. Lai, W. He, X.-L. Zhao, and J. Zeng, "Hyperspectral compressive snapshot reconstruction via coupled low-rank subspace representation and self-supervised deep network," *IEEE Trans. Image Process.*, vol. 33, pp. 926–941, 2024.

[36] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev. Soc. Ind. Appl. Math.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.

[37] J. Liu, P. Musialski, P. Wonka, and J. Ye, "Tensor completion for estimating missing values in visual data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 208–220, Jan. 2013.

[38] E. Kernfeld, M. Kilmer, and S. Aeron, "Tensor-tensor products with invertible linear transforms," *Linear Algebra Appl.*, vol. 485, pp. 545–570, Nov. 2015.

[39] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, and S. Yan, "Tensor robust principal component analysis with a new tensor nuclear norm," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 4, pp. 925–938, Apr. 2020.

[40] M. E. Kilmer, K. Braman, N. Hao, and R. C. Hoover, "Third-order tensors as operators on matrices: A theoretical and computational framework with applications in imaging," *SIAM J. Matrix Anal. Appl.*, vol. 34, no. 1, pp. 148–172, 2013.

[41] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[42] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[43] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Aug. 2002.

[44] B. Le Saux, N. Yokoya, R. Hansch, and S. Prasad, "2018 IEEE GRSS data fusion contest: Multimodal land use classification [technical committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 1, pp. 52–54, Mar. 2018.

[45] M. Simões, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[46] A. Arienzo, G. Vivone, A. Garzelli, L. Alparone, and J. Chanussot, "Full-resolution quality assessment of pansharpening: Theoretical and hands-on approaches," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 168–201, Sep. 2022.

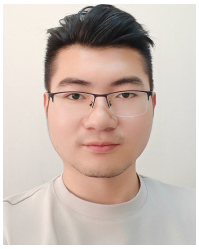
[47] Z.-C. Wu, G. Vivone, T.-Z. Huang, and L.-J. Deng, "Pansharpening via semi-framelet-guided sparse reconstruction," *Inf. Fusion*, vol. 106, Jun. 2024, Art. no. 102297.

[48] Y. Chen, J. Zeng, W. He, X.-L. Zhao, and T.-Z. Huang, "Hyperspectral and multispectral image fusion using factor smoothed tensor ring decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515417.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.

[49] C.-W. Sun, T.-Z. Huang, T. Xu, and L.-J. Deng, "NF-3DLogTNN: An effective hyperspectral and multispectral image fusion method based on nonlocal low-fibered-rank regularization," *Appl. Math. Model.*, vol. 118, pp. 780–797, Jun. 2023.

[50] C. Lu, J. Feng, Z. Lin, and S. Yan, "Exact low tubal rank tensor recovery from Gaussian measurements," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 2504–2510.



Zhong-Cheng Wu received the B.S. degree in information and computing science from Anhui University of Finance and Economics (AUFE), Bengbu, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu, China.

From 2023 to 2024, he was a Visiting Ph.D. Student for statistical signal processing with the National Research Council (CNR), Tito, Italy. His research interests include image processing, tensor learning, and deep learning.



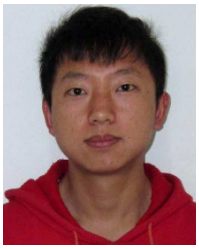
Ya-Jun Li received the B.S. and M.S. degrees in applied mathematics from the School of Mathematical Sciences, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2020 and 2023, respectively.

His research interests include multispectral image fusion and spectral super-resolution.



Ting-Zhu Huang (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in computational mathematics from the Department of Mathematics, Xi'an Jiaotong University, Xi'an, China, in 1986, 1992, and 2001, respectively.

He is currently a Professor with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu, China. His research interests include scientific computation and applications, numerical algorithms for image processing, numerical linear algebra, preconditioning technologies, and matrix analysis with applications.



Liang-Jian Deng (Senior Member, IEEE) received the B.S. and Ph.D. degrees in applied mathematics from the School of Mathematical Sciences, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2010 and 2016, respectively.

From 2013 to 2014, he was a joint-training Ph.D. Student with Case Western Reserve University, Cleveland, OH, USA. In 2017, he joined Hong Kong Baptist University (HKBU), Hong Kong, as a Post-Doctoral Researcher. In addition, he has stayed at the Isaac Newton Institute for Mathematical Sciences, University of Cambridge, Cambridge, U.K., and HKBU, for short visits. He is currently a Professor with the School of Mathematical Sciences, UESTC. His research interests include the use of partial differential equations (PDEs), optimization modeling, and deep learning to address several tasks in image processing, and computer vision, e.g., resolution enhancement and restoration.



Gemine Vivone (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees (summa cum laude) and the Ph.D. degree in information engineering from the University of Salerno, Fisciano, Italy, in 2008, 2011, and 2014, respectively.

He is currently a Researcher with the Institute of Methodologies for Environmental Analysis (IMAA), National Research Council (CNR), Tito, Italy, and also with the National Biodiversity Future Center (NBFC), Palermo, Italy. His main research interests include statistical signal processing, detection of remotely sensed images, data fusion, and tracking algorithms.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:35:14 UTC from IEEE Xplore. Restrictions apply.